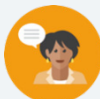


11.2 Subtracting Rational Numbers

Lesson Introduction

 Benchmark	MA.7.NSO.2.2 Add, subtract, multiply, and divide rational numbers with procedural fluency.		 Learning Target	I can fluently subtract rational numbers.
 Benchmark Coherence	Building On		Working Toward	
	MA.6.NSO.2.1 Multiply and divide positive multi-digit numbers with decimals to the thousandths, including using a standard algorithm with procedural fluency. MA.6.NSO.2.2 Extend previous understanding of multiplication and division to compute products and quotients of positive fractions by positive fractions, including mixed numbers, with procedural fluency.		MA.8.NSO.1.7 Solve multi-step mathematical and real-world problems involving the order of operations with rational numbers including exponents and radicals.	
 Essential Questions	- How can number lines be used to subtract rational numbers? - How does the process of subtracting integers apply to subtracting rational numbers?			
 Lesson Narrative	Lesson 2 demonstrates how to find the difference between two numbers by using a variety of strategies, including a number line. Students consider the impact on order in subtraction expressions.			
 Component Summary	11.2.1 Warm-Up (~5 min.)	Analysis of student reasoning for greater fraction (<i>SE p. 171</i>)	11.2.4 Guided Instruction (5 min.)	Exploring the impact of the order when finding the difference between two values (<i>SE p. 174</i>)
	11.2.2 Guided Instruction (5-10 min.)	Finding the difference between two values using number lines (<i>SE p. 172</i>)	11.2.5 Collaboration (10-15 min.)	Finding the difference between two values (<i>SE p. 175</i>)
	11.2.3 Collaboration (10-15 min.)	Using number lines to find the difference between two values (<i>SE p. 173</i>)	11.2.6 Your Turn! (5-10 min.)	Practice finding the difference between two values (<i>SE p. 175</i>)
	11.2.7 Wrap-Up (~5 min.)	Self-reflection on lesson mistake (<i>SE p. 176</i>)		
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> - N/A <u>Additional Terms:</u> - N/A		(Optional) Number Lines	Students may benefit from having number lines that are larger than the ones in the lesson.

 Teacher Tips	Common Misconceptions	Things to Consider
	Students may misunderstand that the value of the numbers to the left of the zero on a number line are decreasing even though the digits are increasing.	- For 11.2.3 Collaboration, assist students with making the connection between the direction used in the process and the sign used in the answers. - If time is an issue, consider assigning 11.2.6 Your Turn! as homework.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Compare and Connect This lesson provides various strategies for the students to use to solve the problems. Prior to completing the 11.2.5 Collaboration, encourage students to compare and connect the different strategies. Allow them to review the strategies and examples from the lesson so that they have a strong understanding of how to execute each strategy before they work together to solve problems.	MA.K12.MTR.3.1: Mathematical Fluency Mathematical Fluency covers selecting effective and applicable methods for solving problems. By presenting students with a variety of strategies and processes, this lesson allows students to choose effective procedures to fluently subtract rational numbers.
 Differentiate Instruction	This lesson has differentiation built in throughout each section. Students use number lines to model subtraction of two rational numbers, providing a visual entry point for learners to engage with the content. Encourage students to use the visual of the number line as needed throughout this lesson to provide an additional support as they build conceptual understanding of adding and subtracting rational numbers.	
Lesson Notes:		

11.2.1 Warm-Up: Math Talk

Component Overview: Two fractions and four student discussions about which fraction is greater are presented. Students will explain which student's reasoning they agree with and will explain why they disagreed with one student's reasoning.



11.2.1 Warm-Up: Math Talk

Several students discussed which fraction was greater, $\frac{13}{19}$ or $\frac{2}{3}$. A summary of the discussion is shown.

Amy:	I changed $\frac{13}{19}$ to $\frac{13}{18}$, and since $\frac{2}{3}$ is equal to $\frac{12}{18}$, I'm about 70% confident that $\frac{13}{19}$ is greater than $\frac{2}{3}$.
Brandon:	The fractions were small enough that I could do mental math and change them into common denominators, so $\frac{13}{19} = \frac{39}{57}$, and $\frac{2}{3} = \frac{38}{57}$. Therefore, $\frac{13}{19}$ is greater than $\frac{2}{3}$.
Carlos:	I changed $\frac{13}{19}$ to $\frac{14}{20}$. Then I thought of a pizza, and that's a lot of little pieces. So, I'm about 70% confident that $\frac{2}{3}$ is bigger than $\frac{13}{19}$.
Dimitri:	I know $\frac{2}{3}$ is $\frac{12}{18}$ if I multiply the top and bottom by 6. $\frac{12}{18}$ is bigger than $\frac{13}{19}$. I'm 90% confident.



The Warm-Up provides an extensive amount of information and a variety of fractions. In order to assist the students with comprehending all of the given information, provide an alternative way to organize the given information such as a three-column table:

Name	Discussion	Answer

- Explain which student's reasoning you agree with.
Answers will vary. I agree with Amy because she compared common denominators. I agree with Brandon because he used the correct process. I agree with Carlos because he chose the bigger numbers. I agree with Dimitri because a smaller numerator could be a bigger fraction.
- Choose one student you didn't agree with, and explain why you did not agree.
Answers will vary. I disagree with Amy because she did not convert correctly. I disagree with Brandon because it was more difficult for me to use mental math like he did. I disagree with Carlos because he did not use a math process to get his answer. I disagree with Dimitri because he did not compare common denominators.

11.2.2 Guided Instruction: Understanding Differences

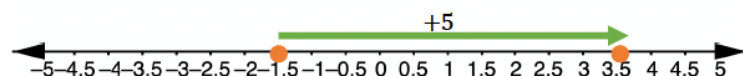
Component Overview: Students will find the difference between two rational numbers using number lines and will complete statements explaining the process.



11.2.2 Guided Instruction: Understanding Differences

Finding the **difference** of two numbers means to *subtract the numbers*. When finding the difference between two numbers, subtract the numbers in the order they are named. For example, the difference of 3.5 and -1.5 is represented by the subtraction expression $3.5 - (-1.5)$.

The difference of two numbers is how far apart the numbers are on a number line.



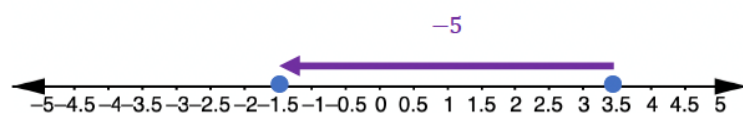
- The numbers 3.5 and -1.5 are 5 units apart. The direction to move from the second number (-1.5) to the first number (3.5) is to the ☐ left, ☒ right, or the ☐ positive ☒ negative direction.

Therefore $3.5 - (-1.5) = +5$.

- Suppose the numbers are subtracted in the opposite order, $(-1.5) - 3.5$.

The numbers 3.5 and -1.5 are still 5 units apart, but the direction of the move has changed. To move from the second number (3.5) to the first number (-1.5) is to move to the ☒ left ☐ right or the ☐ positive ☒ negative direction.

Therefore, $(-1.5) - 3.5 = -5$.



The lesson does not focus on a process. Instead, it is meant to build so the process makes sense for students. Do not reveal any rules until 11.2.4 Guided Instruction.

Students may state that bigger digits actually mean smaller values for the numbers on the left of the zero. Students may also point out that the numbers get smaller going to the left of zero, and bigger going to the right of zero.



In 11.2.2 Guided Instruction, students are provided with number lines to subtract rational numbers. By allowing the students to problem solve with number lines, students are able to visually observe the concept as well as the process for problem solving.

11.2.3 Collaboration: Understanding Differences

Component Overview: With a partner, students will write subtraction expressions and will represent those expressions on number lines. Students will also practice finding the difference between values and will note the direction used to find those differences.



11.2.3 Collaboration: Understanding Differences

- Working with your partner, write each phrase as a subtraction expression. Then, draw a number line representation to find the value of the expression. The first phrase has been done for you.

Phrase	Subtraction Expression	Number Line Representation
The difference of $-\frac{1}{5}$ and $-\frac{4}{5}$	$-\frac{1}{5} - (-\frac{4}{5})$	
Value = $\frac{3}{5}$		

Phrase	Subtraction Expression	Number Line Representation
The difference of $-\frac{4}{5}$ and $-\frac{1}{5}$	$-\frac{4}{5} - (-\frac{1}{5})$	
Value = $-\frac{3}{5}$		

Phrase	Subtraction Expression	Number Line Representation
The difference of 0.9 and -0.5	$0.9 - -0.5$	
Value = 1.4		



Does order matter when translating from verbal to algebraic descriptions? Why or why not?



Monitor for common misconceptions.

- Students may reverse the order of terms.

$$-\frac{1}{5} - (-\frac{4}{5}) = \frac{3}{5}$$

- Students may subtract the second term as positive.

$$-\frac{4}{5} - \frac{1}{5} = -\frac{5}{5}$$

If students make one of these errors, play “My Favorite Mistake” to encourage a growth mindset.

11.2.3 Collaboration: Understanding Differences (continued)

Phrase	Subtraction Expression	Number Line Representation
The difference of -0.5 and 0.9	$-0.5 - 0.9$	
Value = -1.4		

2. With your partner, complete the table to find the difference of the values.

Expression	Distance Between the Values	Direction from Second Value to First Value	Difference
$9.9 - 6.2$	3.7	Right/positive	$9.9 - 6.2 = 3.7$
$9.9 - (-6.2)$	16.1	Right/positive	$9.9 - (-6.2) = 16.1$
$(-9.9) - 6.2$	16.1	Left/negative	$(-9.9) - 6.2 = -16.1$
$(-9.9) - (-6.2)$	3.7	Left/negative	$(-9.9) - (-6.2) = -3.7$
$6.2 - 9.9$	3.7	Left/negative	$6.2 - 9.9 = -3.7$
$6.2 - (-9.9)$	16.1	Right/positive	$6.2 - (-9.9) = 16.1$
$(-6.2) - 9.9$	16.1	Left/negative	$(-6.2) - 9.9 = -16.1$
$(-6.2) - (-9.9)$	3.7	Right/positive	$(-6.2) - (-9.9) = 3.7$



For question 2, the students must identify the correct direction used to reach the answer. Students must also record the correct positive or negative sign on each answer. Pose questions so that students make the connection between the direction used in the process and the sign of the answers, such as:

- When the direction from the second value to the first is left, what is the sign?
- When the direction from the second value to the first is positive, what is the sign?
- Would the association between the direction in the process and the sign of the answer change if the values were on opposite sides of zero?

11.2.4 Guided Instruction: Finding Differences

Component Overview: Students summarize what they learned with number lines to understanding why subtracting a negative value results in plus a positive.



11.2.4 Guided Instruction: Finding Differences

Adding a negative value has the same result as subtracting a positive value. This is because addition and subtraction have an inverse relationship.

For example, $5 + (-6)$ is equivalent to $5 - 6$. Both expressions have a value of -1 .

This relationship can be used to subtract rational numbers. The difference of two numbers can be rewritten as an equivalent addition expression. Then, the rules for addition of two numbers can be followed.

1. Fill in the blanks to complete the table.

Original Expression	$1\frac{2}{5} - \left(-\frac{6}{7}\right)$
Rewrite as an equivalent addition problem by keeping the first term, changing subtraction to addition, and changing the second term to its opposite.	$1\frac{2}{5} \quad \boxed{+} \quad \left(\boxed{+} \frac{6}{7}\right)$
Solve the problem using the rules of addition.	$\boxed{2\frac{9}{35}}$



In order to assist students with correctly applying the rules for subtraction, provide an anchor chart of rational number addition rules and ask students questions about the processes, such as:

- What is the process when the signs are the same?
- What is the process when the signs are different?

11.2.5 Collaboration: Does the Order Matter?

Component Overview: With a partner, students will evaluate subtraction expressions and explain the relationship between two expressions where the order was changed.



11.2.5 Collaboration: Does the Order Matter?

1. With your partner, determine who will be Partner A and who will be Partner B. Partner A will work the problems in Column A while Partner B works the problems in Column B. Then, work together to answer the questions that follow.

Column A	Value	Column B	Value
$3 - 2$	1	$2 - 3$	-1
$5 - (-9)$	14	$(-9) - 5$	-14
$(-11) - 2$	-13	$2 - (-11)$	13
$(-\frac{1}{6}) - (-\frac{1}{3})$	$\frac{1}{6}$	$(-\frac{1}{3}) - (-\frac{1}{6})$	$-\frac{1}{6}$
$(-1.2) - (-3.6)$	2.4	$(-3.6) - (-1.2)$	-2.4
$(-2\frac{1}{2}) - (-3\frac{1}{2})$	1	$(-3\frac{1}{2}) - (-2\frac{1}{2})$	-1

2. Explain the difference between the expressions in Column A and the expressions in Column B.
Answers will vary. The difference in the expressions is that the numbers are written in the reverse order of each other.

3. Explain the relationship between the values of the expressions in Column A and the value of the expressions in Column B.
Answers will vary. The relationship between the values is that they are opposites.



As you circulate note where students determine the pattern in the answers. If students saw the pattern and skipped determining the values of the last three expressions, then determine if they did that because they did not want to find those values or if they understood the concept. Ask students questions such as: How can you use a number line to explain what is happening between the two columns?

11.2.6 Your Turn!

Component Overview: Students work independently to find the difference between two rational numbers.



11.2.6 Your Turn!

1. What is the difference of $\frac{9}{14}$ and $-\frac{13}{20}$?

$$1\frac{41}{140}$$

2. Find the difference for each expression.

a. $5\frac{7}{8} - (-2\frac{2}{5})$

$$8\frac{11}{40}$$

b. $-45.6 - (-1.772)$

$$-43.828$$



Monitor for students who may add or subtract incorrectly. For example:

- Students may reverse the order of terms.

$$-\frac{13}{20} - \frac{9}{14} = -\frac{181}{140}$$

- Students may subtract the second term as positive.

$$\frac{9}{14} - \frac{13}{20} = -\frac{1}{140}$$

If students make one of these errors, play “My Favorite Mistake” to encourage a growth mindset.



Allow the students to associate one subtraction expression or answer with their own appropriate real-life example, such as:

- Ken stood at $-\frac{13}{20}$ feet behind the starting line, jumped $1\frac{41}{140}$ feet over the starting line, and landed $\frac{9}{14}$ feet behind the starting line.

11.2.7 Wrap-Up: Growth Mindset

Component Overview: Students will complete a self-reflection on a mistake they made in the lesson and how they can learn from it.



11.2.7 Wrap-Up: Growth Mindset

Making mistakes is one of the best ways to learn. Reflecting on our mistakes helps our brains to look for that mistake again and to avoid it the next time.

1. What mistake did you make today that you can learn from?

Means

I

Start

To

Acquire

Knowledge

Experience

Skills

Answers will vary. The mistake I made today is that I sometimes moved in the wrong direction on the number line when subtracting rational numbers.



Allow the students to elaborate about their mistakes. Provide sentence stems, such as:

I made a mistake when I _____ because _____.

I should have _____ instead of _____.

The next time I see _____, I will _____.